

B. Another Divisibility Problem

time limit per test: 1 second
memory limit per test: 256 megabytes

Alice and Bob are playing a game in which Alice has given Bob a positive integer $x < 10^8$.

To win the game, Bob has to find another positive integer $y < 10^9$ such that $x \# y$ is divisible by $x + y$.

Here $x \# y$ denotes the integer formed by **concatenating** the integers x and y in that order. For example, if $x = 835$, $y = 47$, then $x \# y = 83547$.

However, since Bob is dumb, he is unable to find such an integer. Please help him.

It can be shown that such an integer always exists.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The only line of each test case contains a single integer x ($1 \leq x < 10^8$) — the integer that Alice has given to Bob.

Output

For each test case, print a single integer y ($1 \leq y < 10^9$) so that Bob can win the game.

If there are multiple answers, print any one of them.

Example

input	Copy
6 8 42 1000 66666 106344 9876543	
output	Copy
1 12 998 7872 8190 174036	

Note

For the first test case, $x = 8$, we can choose $y = 1$, and we have $x \# y = 81$, which is divisible by $x + y = 9$.

For the second test case, $x = 42$, we can choose $y = 12$, and we have $x \# y = 4212$, which is divisible by $x + y = 54$.

Codeforces Round 1049 (Div. 2)

Contest is running

00:36:38

Contestant



→ Submit?

Language:

Python 3.13.2

Almost always, if you send a solution on PyPy, it works much faster

Choose file:

Choose File

 No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	334
Problem B	668
Problem C	835
Problem D	1169
Problem E1	1169
Problem E2	668
Problem E	1837
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 01:23 from the first attempt

